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Self-Reflection Augmented In-Context Learning: Enhancing Machine Translation for China's Minority Languages

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SUMMARY
Large language models (LLMs) have achieved competitive performance in machine translation (MT) via in-context learning (ICL), yet the selection of in-context examples remains an open question that significantly affects translation quality. Existing heuristic-based selection methods lack a targeted mechanism to address translation errors like omissions or mistranslations, which are particularly severe for China's minority languages due to their unique linguistic characteristics, limited resources, and distinct structural challenges. This paper proposes a self-reflection mechanism to optimize example selection for LLM-based MT with the explicit goal of correcting such errors, named Self-Reflection for In-Context Example Selection (SR-ICES). Our approach comprises four sequential stages: (1) an LLM generates an initial translation via a task-specific prompt; (2) the LLM conducts self-reflection to identify erroneous source segments using its internal linguistic knowledge; (3) these erroneous segments are used as a query to retrieve top-N lexically relevant source-target pairs from a parallel corpus; (4) finally, the retrieved examples are fed into the identical LLM again to generate refined translations. Comprehensive experiments on three China's minority language MT tasks, i.e. Uyghur-Chinese (Uy-Zh), Tibetan-Chinese (Ti-Zh), and Mongolian-Chinese (Mn-Zh), show that SR-ICES consistently outperform several competitive baselines, and verify the effectiveness of leveraging LLMs' self-reflection capabilities for in-context example selection, providing a scalable and language-agnostic solution to boost MT performance by closing the loop between translation generation and example selection.
key words: *LLM, In-Context Learning, Self-Reflection, China's Minority Languages*

1. Introduction

Large language models (LLMs) have demonstrated promising performance in machine translation (MT) via in-context learning (ICL)—a paradigm where models adapt to translation tasks through given examples without task-specific fine-tuning[1][2]. However, the effectiveness of ICL heavily relies on the way of choosing in-context examples, which remains an open question. Current approaches to example selection in ICL primarily can be categorized as heuristic based methods, such as random sampling from development datasets or keyword-based retrieval. Such methods fail to systematically address translation errors like omissions or mistranslations, and become even more serious in low-resource or out-of-domain conditions. China's minority languages such as Uyghur, Tibetan, and Mongolian exhibit distinct linguistic features including agglutinative morphology, non-Latin writing systems, flexible word order, and limited parallel corpora, which further amplify translation errors and make conventional ICL strategies less effective. For instance, Agrawal et al.[2] demonstrated that noisy or irrelevant examples can degrade translation quality by over 20% in out-of-domain scenarios—an issue particularly serious in low-resource settings like China's minority language MT.

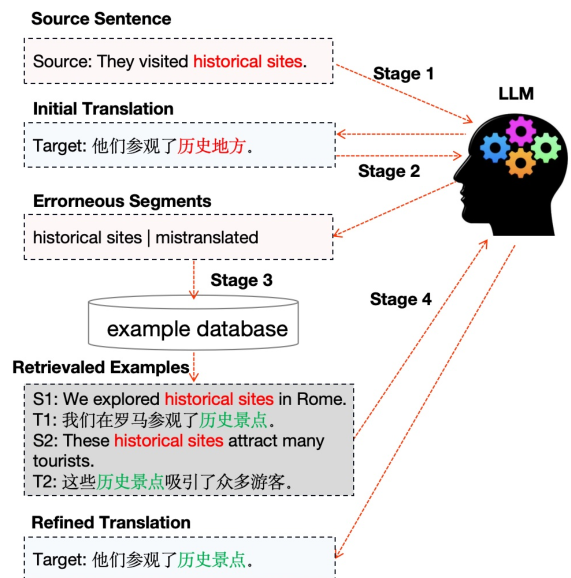

Fig. 1 The overall architecture of our proposed method (Self-Reflection for In-Context Example Selection, SR-ICES). For the sake of readability, we take English-Chinese translation as an example without losing generality. The method consists four sequential stages: initial translation generation, identifying erroneous segments with LLM's self-reflection mechanism, error-driven example retrieval and translation refinement. The red color denote the erroneous segments in the source and target sentence, and the green color denote the correct translation.

To address this question, we proposes a self-reflection mechanism to optimize example selection for LLM-based MT with the explicit goal of correcting such errors, named Self-Reflection for In-Context Example Selection (SR-ICES). As shown in Fig. 1, SR-ICES operates in four sequential stages: (1) generating an initial translation using an LLM via a task-specific prompt; (2) conducting self-reflection to

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pinpoint erroneous source segments (including untranslated and mistranslated content) by leveraging the LLM's internal linguistic knowledge; (3) using these error segments as a query to retrieve top-N lexically relevant source-target pairs from a parallel corpus via the BM25 algorithm; (4) feeding the retrieved examples into a second-round LLM prompt to generate refined translations. This closed-loop framework (initial translation→error detection with self-reflection→example retrieval→refined translation) retrieves examples tailored to the erroneous segments identified in the initial translation, instead of using generic heuristic-based selection strategies. Compared with BM25 ICL that uses the full source sentence as the retrieval query, our method only employs erroneous segments detected via self-reflection to construct queries. This ensures retrieved examples focus on the specific terms and patterns that caused mistranslations or omissions, rather than globally similar but less relevant content.

Experiments are conducted on three China's minority language MT tasks: Uyghur-Chinese (Uy-Zh), Tibetan-Chinese (Ti-Zh), and Mongolian-Chinese (Mn-Zh). Experiments show that our self-reflection mechanism consistently outperform several competitive baselines.

In summary, our work contributes to the growing literature on LLM-based MT by demonstrating that integrating self-reflection into ICL pipelines can significantly improve translation performance. Contrasting to prior studies that rely on static or human-curated examples, our approach dynamically adapts to translation errors, which offers a scalable and language-agnostic solution for ICL in LLM based machine translation.

2. Methods

Our proposed method (Self-Reflection for In-Context Example Selection, SR-ICES) is composed of four sequential stages: initial translation generation, self-reflection, error-driven example retrieval and translation refinement, as is introduced in details in the following subsections respectively.

2.1 Initial Translation Generation

The first stage involves generating an initial translation using the LLM (e.g., Qwen). Formally, given a source sentence x, we first construct a prompt P_{init} that contains a general machine translation instruction. The LLM then generates an initial translation $\hat{y}_{\text{init}}$ conditioned on P_{init} and x. Concretely, the prompt is structured to include clear instructions, as follows:

> "Translate the following [source_language] sentence to [target_language] accurately.
> Source: [input_source_sentence]
> Target Translation: "

where `[source_language]` (`[target_language]`) is replaced with the specific source (target) language, and `[input_source_sentence]` is the test sentence to be translated.

2.2 Error Detection with Self-Reflection

The initial translation $\hat{y}_{\text{init}}$ is fed back to the identical LLM with a self-reflection prompt P_{reflect}. This prompt instructs the LLM to compare $\hat{y}_{\text{init}}$ against the source x and detect specific segments where the translation is possibly erroneous—including **untranslated segments** (parts of x with no corresponding output in $\hat{y}_{\text{init}}$) and **mistranslated segments** (parts of x with semantically or grammatically mistranslated in $\hat{y}_{\text{init}}$).

The detection is based on the internal linguistic knowledge and cross-lingual alignment capabilities of the LLM . During the self-reflection stage, the model is prompted to act as a "critic" rather than a "translator," comparing the source and target to find semantic discrepancies. Even if a model's generation is flawed, its discriminative capability is often stronger, allowing it to identify inconsistent meanings between the source and its own initial output. Due to the principle, the LLM can generate a series of error segments $E = \{e_1, e_2, \ldots, e_k\}$ from the source sentence with sophisticated designed prompts. In experiments, we configure the prompt P_{reflect} as:

> "Analyze the following [source_language]-[target_language] translation pair. Identify all source segments with critical translation errors. For each error, provide: (1) the exact segment text from the source sentence; (2) the error type (untranslated or mistranslated). Ignore trivial issues (e.g., punctuation).
> Source: [input_source_sentence]
> Initial Translation: [y_init]
> Error Segments (format: Segment — Error Type):"

2.3 Error-Driven Example Retrieval

We connect each identified error segment $e_i \in E$ as a single query e to retrieve relevant examples from a parallel corpus $\mathcal{D}$. We employ the BM25 retrieval algorithm, which ranks parallel sentences based on lexical overlap between the query e and source sentences in $\mathcal{D}$. BM25 is chosen for its efficiency and unsupervised nature, emphasizing lexical matching without requiring training. The BM25 score between e and a candidate source sentence x_c is calculated as:

$$\text{BM25}(e, x_c) = \sum_{t \in e} \left(\frac{(k_1 + 1) \cdot f(t, x_c)}{f(t, x_c) + k_1 \cdot \left(1 - b + b \cdot \frac{|x_c|}{\text{avgdl}}\right)} \right) \tag{1}$$

$$\cdot \log \left(\frac{N(\mathcal{D}) - n(t) + 0.5}{n(t) + 0.5} \right) \tag{2}$$

where $f(t, x_c)$ is the term frequency of token t in x_c, $|x_c|$ is the length of x_c, avgdl is the average document length in $\mathcal{D}$, $N(\mathcal{D})$ is the total number of documents in $\mathcal{D}$, and $n(t)$ is the number of documents containing token t. We keep the top-N most relevant source–target pairs retrieved, noted as D_{retrieve}.

2.4 Translation Refinement

The final stage aims to improve the translation quality with the help of retrieved relevant examples in the last stage. A new prompt P_{refine} is constructed by prepending D_{retrieve} to the original source x. The LLM then generates a refined translation $\hat{y}_{\text{final}}$ conditioned on P_{refine}, as below:

> "Translate the following [source_language] sentence to [target_ language] accurately. The examples below address critical translation errors (including untranslated and mistranslated segments) present in the source sentence, use them to refine your initial translation.
> ===================
> Examples (Source → Target):
> 1. [source_1] → [target_1]
> 2. [source_2] → [target_2]
> ...
> N. [source_N] → [target_N]
>
> Source Sentence: [input_source_sentence]
> Initial Translation: [init_translation]
> Refined Translation:"

where each `[source_i]` and `[target_i]` is the i-th source-target pair from the top-N examples, which possible contains the correct translation for the erroneous segments detected in stage 2.

This four-stage pipeline enables the LLM to self-diagnose translation errors and dynamically retrieve in-context examples that oriented for those errors, thereby closing the loop between translation generation and example selection.

3. Experiments

3.1 Data and Setup

To evaluate the effectiveness of our proposed method, we conduct comprehensive experiments on three Chinese minority language MT tasks: Uyghur-Chinese (Uy-Zh), Tibetan-Chinese (Ti-Zh), and Mongolian-Chinese (Mn-Zh). The experiments aim to answer our two core concerns: (1) Does SR-ICES outperform traditional in-context learning (ICL) baselines in translation quality? (2) How does the number of retrieved examples affect the machine translation performance?

For evaluation, we adopt 2 kinds of parallel corpus released by CCMT2025† as the example database and test set for each translation task. In concrete, we use XJIPC-test-uyzh-CWMT2018, QHNU-test-tizh-CWMT2018, and IMU-test-mnzh-CWMT 2018 as test set for Uyghur-Chinese (Uy-Zh), Tibetan-Chinese (Ti-Zh), and Mongolian-Chinese (Mn-Zh) translation respectively, and each of them contains 1,000 parallel sentence pairs. We adopt the parallel corpus contained in the train set released by CCMT2025 to construct example database, the number of sentences are 170057, 156578 and 1259772 for Uy-Zh, Ti-Zh, and Mn-Zh, respectively.

For the BM25 algorithm, we set $k_1 = 1.5$, $b = 0.75$, and N as 10 for all the tasks. Since BM25 relies on word-level similarity, both source and target sentences need segmentation or tokenization. For Tibetan, the *pybo* tokenizer is adopted, which is a tool specifically designed for Tibetan text segmentation. For Chinese, *jieba* tokenizer is used, which is a widely adopted tool for Chinese word segmentation. For Uyghur and Mongolian, we simply tokenize them with space delimiters.

3.2 Results of Machine Translation

We compare our proposed SR-ICES with three other LLM-based MT methods, as following:

- **LLM**: Direct zero-shot translation by LLM without in-context examples.
- **Random ICL**: Randomly selects N in-context examples from the training set.
- **BM25 ICL**: Uses the entire source sentence as a BM25 query to retrieve Top-N examples without self-reflection.
- **TasTe**[3]: A state-of-the-art self-reflection based translation method that refines translations via self-correction without retrieval-augmented example selection.

To ensure a fair comparison, all the large language models employed in this paper are Alibaba's Qwen3††, with the specific version qwen-plus-2025-07-28. This model is chosen for its strong cross-lingual capabilities, particularly in Chinese and China's minority languages, and its efficiency in processing translation tasks with convenient cloud API access.

We use char-based BLEU-4 as the evaluation metric[4]. To ensure the reliability and stability of experimental results, all experiments are independently repeated 5 times, and the average BLEU scores are reported with statistical significance tested ($p < 0.05$)[5]. As shown in Table 1, BM25 ICL achieves the most significant improvement margins over all the translation tasks, compared to the vanilla LLM and Random ICL, which is consistent with previous research[2], further validating that in-context learning is indispensable for improving low-resource language transla-

†http://mteval.cipsc.org.cn:81/CCMT2025/index.html
††https://www.qianwen.com/

Table 1 BLEU Scores Evaluated on Test Set

Method	Uy-Zh	Ti-Zh	Mn-Zh	Avg.
LLM	34.78	11.82	3.15	16.58
Random ICL	40.24	13.22	5.52	19.66
BM25 ICL	45.55	45.16	29.38	40.03
TasTe	36.92	15.73	9.61	20.75
SR-ICES	**47.32**	**46.90**	**29.96**	**41.39**

tion. Building on this, our proposed SR-ICES advances performance by introducing an error-driven example selection mechanism—instead of relying on full-sentence like BM25 ICL, it first identifies erroneous segments via LLM self-evaluation, then retrieves examples targeted at these specific weaknesses. This design enables SR-ICES to achieve the highest average BLEU (41.39), outperforming BM25 ICL (40.03) by 1.36 points on average. Language-specifically, compared to BM25 ICL, SR-ICES achieves 47.32 (+1.77) for Uy-Zh, 46.90 (+1.74) for Ti-Zh, and 29.96 (+0.58) for Mn-Zh, respectively. Moreover, SR-ICES also outperforms the advanced self-reflection method TASTE by 20.21 points on average, further demonstrating the superiority of using retrieves examples.

From the overall results, we can observe that all in-context learning based methods significantly outperform the zero-shot LLM baseline, which confirms the importance of using demonstration examples for low-resource minority language translation. BM25 ICL achieves substantial improvements by retrieving semantically similar examples, while TASTE, as a pure self-refinement method without in-context demonstrations, yields relatively low scores. This indicates that self-reflection alone is insufficient to address the challenges of scarce data and large linguistic divergence. In contrast, our SR-ICES consistently achieves the best performance by integrating self-reflection error detection with error-driven in-context example retrieval. The targeted example selection mechanism effectively mitigates mistranslations and omissions, making it more suitable for the three typical Chinese minority language translation tasks.

3.3 Impact of Example Number

To answer the second concern aforementioned, i.e. investigating how the number of in-context examples affects translation quality, we design a controlled experiment with various values for N ranging from 0 to 30. As shown in Fig.2, the translation performance across Uy-Zh, Ti-Zh, and Mn-Zh tasks generally improves with more in-context examples. When the number of examples rises from 0 to 10, the BLEU scores increase rapidly. However, when the number exceeds 10, the growth slows significantly. The explanation is possibly that excessive examples cause redundancy, and finally leading to marginal improvements or slight drops. On the other hand, 10 examples maybe already cover most of the information needed for correcting the translation.

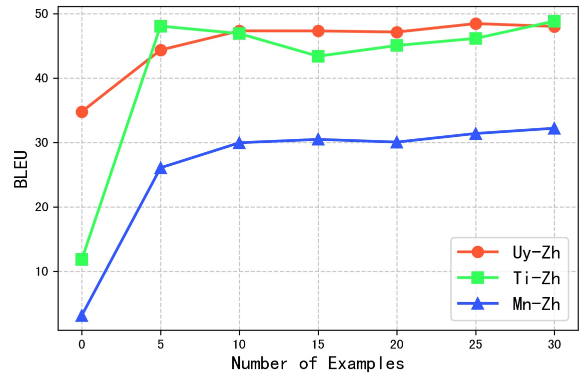

Fig. 2 The BLUE scores with varying example number.

4. Related Work

In-Context Learning for Machine Translation In-context learning has drawn researchers' attentions in the field of LLM based machine translation[2][6]. ICL allows LLMs to perform tasks by conditioning on a few in-context examples provided within the prompt. Early studies demonstrated that LLMs like GPT-4 can achieve competitive performance on high-resource language pairs even using randomly selected in-context examples, but their effectiveness diminishes in low-resource scenarios[2]. For instance, Agrawal et al. showed that a single noisy example in 1-shot ICL could degrade BLEU scores by up to 20% in out-of-domain translations, emphasizing the critical role of example quality. Most prior studies in this area have mainly focused on boosting translation quality by optimizing the selection and presentation of in-context examples. However, the problem of how to systematically select the most relevant and effective in-context examples remains an open challenge. Our work addresses this by introducing a self-reflection mechanism to identify errors in initial translations and then retrieve relevant examples in terms of the error information, which is different from the existing works that often rely on static or heuristic-based example selection methods.

Self-reflection for LLM based Mchine Translation Self-reflection, a process where LLMs evaluate and refine their own outputs, has gained traction for improving LLM performance. Recent works [7][6][3] introduce iterative refinement loops, where LLMs first generate translations, identify errors via automatic metrics (e.g., MQM), and then correct them. These methods demonstrate that feedback-driven refinement can reduce translation errors, particularly in addressing fluency and terminology issues. In contrast to these efforts, our work focuses on example selection rather than direct text correction. By leveraging LLMs to identify erroneous source segments and retrieve contextually relevant examples, we transform the translation task into a retrieval query generation task. This coincides with the "retrieval-augmented generation" (RAG) paradigm, where external knowledge (e.g., parallel corpus) is integrated to address specific weaknesses in the LLM's initial output.

Automatic Post-Editing Automatic post-editing (APE) aims to automatically correct the output of a machine translation system, similar to the role of human post-editors[8][9]. Generally, APE systems require a new model trained using triplets in the format of the source text, the machine-translated text, and the post-edited version of the machine-translated text. In contrast to APE, our self-reflection framework for in-context example selection focuses on improving the translation process using a single LLM model. By integrating self-evaluation and error-driven example retrieval, we aim to guide the LLM to generate a more accurate translation instead of relying on post-hoc correction, thus without the need of making post-edited data for training a new model.

5. Conclusion

This paper addresses the key limitation of traditional in-context learning in LLM based machine translation, that is the lack of goal-oriented example selection. We propose a novel Self-Reflection for In-Context Example Selection (SR-ICES) framework, which empowers LLM to dynamically identify translation weaknesses and retrieve more targeted in-context examples, closing the loop between translation generation and example selection. Comprehensive experiments on three China's minority language MT tasks validate the effectiveness of the proposed method. While our current study is centered on Chinese machine translation, future research includes extending this approach to validate its efficacy across broader language pairs and conducting experiments using varying LLMs.

Acknowledgments

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References

[1] T.B. Brown, B. Mann, and etc., "Language models are few-shot learners," "Proceedings of NeurIPS 2020, Online, p.1877–1901, December 2020.
[2] S. Agrawal, C. Zhou, M. Lewis, L. Zettlemoyer, and M. Ghazvininejad, "In-context examples selection for machine translation," Proceedings of ACL, Toronto, Canada, pp.8857–8873, July 2023.
[3] Y. Wang, J. Zeng, X. Liu, F. Meng, J. Zhou, and M. Zhang, "Taste: Teaching large language models to translate through self-reflection," Proceedings of ACL, Bangkok, Thailand, pp.6144–6158, August 2024.
[4] P. Kishore, R. Salim, T. Ward, and W.J. Zhu, "Bleu: a method for automatic evaluation of machine translation," Proceedings of ACL, Philadelphia, USA, pp.311–318, Association for Computational Linguistics, July 2002.
[5] M. Collins, P. Koehn, and I. Kucerova, "Clause restructuring for statistical machine translation," Proceedings of ACL, USA, pp.531–540, 2005.
[6] C. Zhang, X. Liu, J. Lin, and Y. Feng, "Teaching large language models an unseen language on the fly," Proceedings of ACL, Bangkok, Thailand, pp.8783–8800, August 2024.
[7] Z. Feng, Y. Zhang, H. Li, B. Wu, J. Liao, W. Liu, J. Lang, Y. Feng, J. Wu, and Z. Liu, "Tear: Improving llm-based machine translation with systematic self-refinement," Proceedings of NAACL, Albuquerque, New Mexico, USA, pp.3922–3938, May 2025.
[8] D. Ki and M. Carpuat, "Guiding large language models to post-edit machine translation with error annotations," Proceedings of NAACL, Mexico City, Mexico, pp.4253–4273, June 2024.
[9] Z. Li, J. Ma, and F. Ren, "Exploiting multi-level data uncertainty for japanese-chinese neural machine translation," IEICE Trans. Inf. Syst., vol.108, no.5, pp.440–443, 2025.